

# Physical Units Expression Language

PUEL Specification for SciNumTools v3

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# 1 PUEL Specification

## 1.1 Overview

PUEL (Physical Units Expression Language) is a domain-specific language for representing and manipulating physical units and dimensional expressions.

It defines the syntax and structure of unit expressions used in scientific computations within the SciNumTools3 framework. Evaluation, normalization, dimensional analysis, and unit conversion are performed by the underlying EXS and PUQ modules.

The language is designed for the following use cases: \* As a component of DIPL for defining and validating unit-aware parameters. \* As a standalone language for the concise and machine-readable representation of physical unit expressions. \* As a backend for programmatic interfaces, such as Python bindings and other language integrations.

This chapter defines the language syntax and conversion rules. The following chapters provide reference tables for scaling prefixes, unit systems, and shared definitions, followed by the derived units, constants, and quantity scales of each supported system.

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## 1.2 Design Principles

PUEL is designed to be:

- **Concise** — compact textual representation of units

- **Composable** — units can be combined using algebraic operations
- **Deterministic** — expressions resolve to a canonical dimensional form
- **Extensible** — supports custom units and prefixes
- **Solver-backed** — evaluation delegated to EXS

### 1.2.1 Relationship to Existing Unit Standards

PUEL is designed as a machine-readable language for representing physical quantities and unit expressions, rather than as a replacement for established unit coding systems. In particular, PUEL is intended to be interoperable with standards and vocabularies such as the International System of Units (SI), the Unified Code for Units of Measure (UCUM), UDUNITS, QUDT, and the OM ontology. PUEL extends the scope of conventional unit notations by providing a unified syntax for numerical values, units, uncertainties, physical constants, symbolic quantities, and algebraic expressions. Where applicable, PUEL unit expressions **SHOULD** be convertible to and from established unit representations, while the semantic interpretation of a PUEL expression remains independent of any particular serialization format. This design allows PUEL to serve as a compact interchange notation for scientific quantities while remaining compatible with existing unit standards and ontologies.

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## 1.3 Unit Model

### 1.3.1 Base units

All unit expressions are ultimately represented in terms of a fixed set of base dimensions:

- Length in meters (**m**)
- Mass in grams (**g**)
- Time in seconds (**s**)
- Temperature in kelvins (**K**)
- Electric current in amperes (**A**)
- Luminous intensity in candelas (**cd**)
- Amount of substance in moles (**mol**)
- Angle in radians (**rad**)

Derived units are represented as combinations of these base dimensions together with an associated scaling factor.

The choice of base dimensions is driven by implementation considerations and does not exactly match the SI base-unit system.

- Mass is represented using grams (**g**) rather than kilograms (**kg**) to avoid embedding a prefix in a base unit.
- Electric current is represented using amperes (**A**); electric charge is a derived quantity (**C = A\*s**).
- Angle is included as a base dimension to enable dimensional validation of mathematical functions that require angular arguments. However, angular units **MUST** remain convertible to dimensionless units, reflecting their dimensionless nature in the SI system.

### 1.3.2 Scaling factors

Scaling prefixes provide a compact way to represent decimal multiples and submultiples of units. A scaling prefix modifies a unit identifier by applying a fixed numerical factor while preserving the unit's dimensionality.

For example, the expressions **m**, **cm**, and **km** all represent units of length and differ only in their scaling factors relative to the base unit.

PUEL defines a standard set of scaling prefixes. Implementations **MUST** recognize and correctly apply these prefixes when parsing unit expressions. However, prefixes **SHOULD** only be permitted for units for which their use is conventional or widely accepted.

The complete list of supported scaling prefixes is provided in Scaling Prefixes.

### 1.3.3 Named Entities

Named entities in PUEL include all units (e.g., Joule J, degree Celsius `Cel`, decibel `dB`), physical constants (e.g., proton mass `{m_p}`, Avogadro constant `{N_A}`, solar luminosity `{L_sol}`), physical quantities (e.g., energy `<E>`, electric flux `<Phi_E>`, radiation dose `<D_r>`), and unit-system scaling factors (e.g., `|E|`, `|Phi_E|`, `|D_r|`). The dimensionality of these entities is defined in terms of the base dimensions described in the previous section.

PUEL distinguishes between these categories using the following notation:

- Units are represented by their symbols: J, `Cel`, `dB`
- Physical constants are enclosed in curly braces: `{m_p}`, `{N_A}`, `{L_sol}`
- Physical quantities are enclosed in angle brackets: `<E>`, `<Phi_E>`, `<D_r>`
- Unit-system scaling factors are enclosed in pipe symbols: `|E|`, `|Phi_E|`, `|D_r|`

Unit-system scaling factors represent the numerical coefficients required to convert a physical quantity between the SI system and another system of units. They are primarily intended to support conversions between systems with different dimensional conventions, such as SI and CGS.

Each implementation **MUST** define the numerical values and dimensional representations of all supported derived entities. These definitions **SHOULD** be precomputed and stored to avoid repeated evaluation and dimensional derivation during the processing of unit expressions.

Within a given implementation and unit system, all supported identifiers and scaling-prefix combinations **MUST** be unique. Consequently, every valid token in a PUEL expression **MUST** have exactly one interpretation.

### 1.3.4 Custom units

Custom units **MAY** be defined in the surrounding environment using existing unit expressions. Once registered in a unit system, a custom symbol can be used in expressions just like a built-in unit, including in products, quotients, and powers. For example, an environment may define `step` as `0.75*m` and then interpret `4*step` as `3*m`.

Custom definitions are scoped to their unit system. PUQ stores their numerical scale, uncertainty when present, and dimensions. Registration is performed by the host application; PUEL does not define assignment syntax. PUQ custom units do not accept scaling prefixes.

---

## 1.4 Unit expressions

### 1.4.1 General Form

A unit expression consists of an optional unit system identifier followed by a numerical value, unit factors, or both.

General form:

`[<system>_] [<value>*] [<unit-expression>]`

where:

- `<system>` is a unit system identifier.
- `<value>` is a numerical value, optionally including an uncertainty.
- `<unit-expression>` is a combination of unit identifiers and algebraic operators.

At least one of `<value>` or `<unit-expression>` **MUST** be present.

The `*` separator between the value and unit factors is present only when both are supplied. A value alone, such as `2.5`, represents a dimensionless quantity. The value **MAY** also be a numerical array, as described below.

Examples:

```
m
2.5*m
3.452(3)*kg*m/s2
US_lb*ft
SI_9.81*m/s2
```

The following rules apply in unit expressions:

| Rule                                                                                                                                                                                                                         | Examples                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expressions MUST NOT contain whitespace except for spaces after array separators.                                                                                                                                            | <code>kg*m/s2</code>                                                                                                                                                |
| Unit identifiers MAY be combined using multiplication (*), division (/), and grouping parentheses ((, )).                                                                                                                    | <code>kg*mm/skg*m2/(sr*s2)</code>                                                                                                                                   |
| Exponents are written directly after the corresponding unit identifier and do not use exponentiation operators such as ^ or **.                                                                                              | <code>m2</code>                                                                                                                                                     |
| Fractional exponents are written using the <code>numerator:denominator</code> notation.                                                                                                                                      | <code>kg3:2</code>                                                                                                                                                  |
| Negative exponents are written using a leading minus sign.                                                                                                                                                                   | <code>s-1</code>                                                                                                                                                    |
| A unit system MAY be specified. When present, it MUST appear at the beginning of the expression and MUST be separated from the remainder by an underscore (_).                                                               | <code>US_1b*ft</code>                                                                                                                                               |
| Numerical values MAY be included in expressions.                                                                                                                                                                             | <code>2.34e3*mol-9.81*m/s2</code>                                                                                                                                   |
| The symbols + and - denote the sign of a numerical value or exponent; they do not represent addition or subtraction operators.                                                                                               | <code>+5*m-5*ms-2</code>                                                                                                                                            |
| Uncertainties are written in parentheses immediately following the last decimal digit of a numerical value. The number of digits in the uncertainty corresponds to the same number of least significant digits in the value. | <code>3.45234(2)e3 → 3452.34 ±</code><br><code>0.023.45234(12) → 3.45234 ±</code><br><code>0.0001212.3(4)e-2 → 0.123 ±</code><br><code>0.004120(5) → 120 ± 5</code> |

### 1.4.2 Expression Parsing

Although numerical values are typically written at the beginning of an expression, they are not restricted to that position. Numerical factors MAY appear anywhere within a unit expression as multiplicative constants. For example, `2*kg*m/s2` and `kg*(2*m)/s2` are equivalent.

Implementations SHOULD accept numerical factors in any valid position within a unit expression. However, producers of unit expressions SHOULD place any numerical factor at the beginning of the expression, as this improves readability and simplifies parsing.

Unit expressions MAY use division operators and grouping parentheses for readability. For example, `kg*m2/(sr*s2)` and `kg*m2*sr-1*s-2` represent the same unit expression.

An exponent MAY follow a closing parenthesis and applies to the entire grouped expression, including numerical factors. Integer, negative, and fractional exponents use the same notation as unit exponents. For example, `(m/s)2` is equivalent to `m2*s-2`, and `kg/(m*s)-1:2` is equivalent to `kg*m1:2*s1:2`. Exponents also apply to named entities, as in `kg*<v>2`.

Expressions containing only numerical factors are supported. For example, `3*(2.0e1/5.0)` evaluates to the dimensionless value 12.

### 1.4.3 Numerical arrays

A numerical array is written as a comma-separated list of values in square brackets. All elements share the units specified outside the brackets. Elements MAY include scientific notation or parenthesized uncertainties. Spaces MAY follow commas.

```
[2,3.4,5e6]*km/s
[2.00(20), 3.00(30)]*m
```

Multiplication and division by a scalar apply to each element. Multiplication and division between arrays of the same shape are elementwise operations. For example, `[20,40.5]*2` evaluates to `[40,81]`, and `[20,40.5]*[2,3]` evaluates to `[40,121.5]`. Unit factors are combined using the ordinary expression rules.

### 1.4.4 Unit-system selection

The system identifier, when present, MUST appear only at the beginning of the expression, immediately followed by an underscore (\_). It selects the interpretation of all unit factors in the expression. For example, `US_23*ft2` is valid; `US 23*ft2` and `23*US_ft2` are not valid PUEL expressions. PUQ supports SI and, when enabled, CGS variants (ESU, EMU, GU), English systems (IU, US), and natural systems (AU, PU, SRU, GRU, GEO).

When no identifier is supplied, PUQ uses the active unit system of the surrounding environment, initially SI. If both the expression and its environment explicitly select a system for that expression, the selections MUST

agree. Producers SHOULD include the system identifier when an expression must retain its interpretation outside its original environment.

#### 1.4.5 Display notation

In addition to ASCII notation, PUQ can produce Unicode and HTML representations for display. When Unicode input preprocessing is enabled, it accepts superscript exponents, the Unicode middle dot (U+22C5) for multiplication, the Unicode minus sign (U+2212), and the multiplication sign (U+00D7) for scientific notation. For example, a displayed form using U+00D7 before 10 and superscript U+2212/U+22C5 symbols corresponds to the portable expression `6.23537e-2*C3`.

Unicode input conveniences are optional; ASCII notation remains the portable form. A system identifier MUST retain its leading position and underscore separator even when Unicode notation is used, as in the displayed form of `US_23*ft2`. HTML output is a display format, not an expression input format.

#### 1.4.6 Canonical Representation and Serialization

Implementations SHOULD internally normalize unit expressions to a canonical form in which all unit factors appear in the numerator and division is represented using negative exponents. For example, the expression `kg*m2/(sr*s2)` is canonically represented as `kg*m2*sr-1*s-2`.

The canonical form provides a unique and deterministic textual representation of a unit expression. Because the original placement of division operators and grouping parentheses cannot, in general, be reconstructed from a normalized expression, implementations SHOULD use the canonical form when serializing unit expressions.

If a unit expression contains a numerical factor, serialized expressions MUST place the numerical factor at the beginning of the expression.

When representing base dimensions, only dimensions with non-zero exponents SHOULD be included. The order of the base dimensions MUST be preserved as specified in Section 3.1..

### 1.5 Conversions

PUEL supports conversion between units of identical dimensions:

`km` → `m`  
`eV` → `J`  
`dBW` → `W`

Special handling exists for:

- logarithmic units
- temperature units

These require custom conversion rules.

#### 1.5.1 Logarithmic and temperature conversions

Logarithmic conversions distinguish ratios from levels with a defined reference. PUQ supports power ratios (`PR`), amplitude ratios (`AR`), and logarithmic units such as decibels (`dB`) and nepers (`Np`). Power levels such as `dBm` and `dBW` include their reference power: `0*dBm` corresponds to `1*mW`, while `0*dBW` corresponds to `1*W`. A bare `dB` value does not specify an absolute power.

Absolute temperature conversions can include an offset as well as a scale. For example, `23*Ce1` converts to `296.15*K`. PUQ supports Celsius (`Ce1`), Fahrenheit (`degF`), kelvin (`K`), and Rankine (`degR`). Such conversions cannot be described by a multiplicative scaling factor alone.

#### 1.5.2 Conversion between unit systems

Conversions between systems with different dimensional conventions may require a physical-quantity context in addition to source and target units. For example, converting SI current in `A` to electrostatic CGS current in `statA` requires the current context `I`. Energy conversion between `J` and `erg` does not require an explicit context.

The context identifies the corresponding physical quantity in each system and enables the use of its unit-system scaling factors. It is supplied by the surrounding application, not appended as an additional operator in the expression. A system prefix selects a system; it does not by itself specify a conversion context.

## 2 Scaling prefixes

| Prefix | Factor | Name  |
|--------|--------|-------|
| Y      | 1e24   | yotta |
| Z      | 1e21   | zetta |
| E      | 1e18   | exa   |
| P      | 1e15   | peta  |
| T      | 1e12   | tera  |
| G      | 1e9    | giga  |
| M      | 1e6    | mega  |
| k      | 1e3    | kilo  |
| h      | 1e2    | hecto |
| da     | 1e1    | deka  |
| d      | 1e-1   | deci  |
| c      | 1e-2   | centi |
| m      | 1e-3   | milli |
| u      | 1e-6   | micro |
| n      | 1e-9   | nano  |
| p      | 1e-12  | pico  |
| f      | 1e-15  | femto |
| a      | 1e-18  | atto  |
| z      | 1e-21  | zepto |
| y      | 1e-24  | yocto |

[!NOTE] The prefix u is used to represent micro (Unicode U+03BC) in ASCII-only unit expressions.

## 3 Unit systems

The following tables specify the built-in vocabulary provided by PUQ for PUEL expressions. A system identifier appears only at the beginning of an expression and is immediately followed by `_`, for example `SI_2*J` or `US_2*gal`. An identifier selects the entire expression; it cannot qualify an individual factor.

| Identifier | System                        | Unit and quantity tables |
|------------|-------------------------------|--------------------------|
| SI         | International System of Units | SI definitions           |
| ESU        | Electrostatic CGS units       | ESU definitions          |
| EMU        | Electromagnetic CGS units     | EMU definitions          |
| GU         | Gaussian CGS units            | GU definitions           |
| IU         | Imperial units                | IU definitions           |
| US         | United States customary units | US definitions           |
| AU         | Atomic units                  | AU definitions           |
| PU         | Planck units                  | PU definitions           |
| SRU        | Units for special relativity  | SRU definitions          |
| GRU        | Units for general relativity  | GRU definitions          |
| GEO        | Geometrized units             | GEO definitions          |

### 3.1 Reading the tables

- Every system includes the shared base-unit symbols. Each system page lists its additional units and aliases, constants, and quantity scales. A unit available in SI is not automatically available in another system.
- Definitions describe one unprefix unit in terms of symbols from the same system. Prefix permissions apply to the symbol in that row; prefixes are not implicitly inherited through an alias.
- Numerical factors and constant values reproduce the PUQ definitions, including their stated precision. These tables describe the PUEL vocabulary and do not redefine external metrological standards.
- Temperature and logarithmic rows give reference units, not linear equalities. Use the special conversion rules; for example, `Ce1` is not equal to `K`.
- SI is always available in PUQ. Other system families, offset temperature units, and logarithmic units depend on enabled implementation features.

- Natural systems often use quantity scales such as <E> rather than a new named unit. Constants without # may be normalized to unity; fixed {#...} reference constants retain their reference values.

## 4 Shared unit definitions

These symbols are available in every listed system. Prefixes are case-sensitive: **All** means any prefix in the prefix table, **None** means no prefix, and an explicit list permits only those prefixes. An unprefix symbol is always permitted.

### 4.1 Base-unit symbols

| Symbol | Meaning | Definition or reference | Prefixes |
|--------|---------|-------------------------|----------|
| m      | meter   | m                       | All      |
| g      | gram    | g                       | All      |
| s      | second  | s                       | All      |
| K      | Kelvin  | K                       | All      |
| A      | Ampere  | A                       | All      |
| cd     | candela | cd                      | All      |
| mol    | mole    | mol                     | All      |
| rad    | radian  | rad                     | m        |

### 4.2 Fixed reference constants

The following constants are available in SI, AU, PU, SRU, GRU, and GEO. The # inside the braces distinguishes a fixed reference constant from a system-specific constant: for example, {#c} retains its reference value while {c} may be 1 in a natural system. Parenthesized digits denote uncertainty using the ordinary PUEL notation.

| Symbol   | Meaning           | Definition or reference             | Prefixes |
|----------|-------------------|-------------------------------------|----------|
| {#alpha} | fine str. const.  | 7.2973525643(11)e-3                 | None     |
| {#euler} | Euler's num.      | 2.718281828459                      | None     |
| {#N_0}   | Avogadro's num.   | 6.02214076e23                       | None     |
| {#pi}    | pi num.           | 3.1415926535                        | None     |
| {#a_0}   | Bohr radius       | 5.29177210544(82)e-11*m             | None     |
| {#c}     | speed of light    | 299792458*m/s                       | None     |
| {#e}     | elem. charge      | 1.602176634e-19*A*s                 | None     |
| {#eps_0} | permit. of vac.   | 8.8541878188(14)e-12*m-3*kg-1*s4*A2 | None     |
| {#G}     | grav. const.      | 6.67430(15)e-11*m3*kg-1*s-2         | None     |
| {#g}     | grav. accel.      | 9.80665*m/s2                        | None     |
| {#h}     | Planck const.     | 6.62607015e-34*m2*kg*s-1            | None     |
| {#hbar}  | Reduced Pl. con.  | 1.054571817e-34*m2*kg*s-1           | None     |
| {#k_B}   | Boltzmann const.  | 1.380649e-23*m2*kg*s-2*K-1          | None     |
| {#mu_0}  | permeab. of vac.  | 1.25663706127(20)e-6*m*kg-1*s-2*A-2 | None     |
| {#mu_B}  | Bohr magneton     | 9.2740100657(29)e-24*m2*A-1         | None     |
| {#m_e}   | electron mass     | 9.1093837139(28)e-31*kg             | None     |
| {#m_p}   | proton mass       | 1.67262192595(52)e-27*kg            | None     |
| {#m_n}   | neutron mass      | 1.67492749804(95)e-27*kg            | None     |
| {#R_inf} | Rydberg constant  | 10973731.568157(12)*m-1             | None     |
| {#sigma} | Stef.Bolt. const. | 5.670374419e-8*kg*s-3*K-4           | None     |
| {#N_A}   | Avogadro's const. | 6.02214076e23/mol                   | None     |
| {#kap_0} | permittivity      | 1.11265005620e-13*m-3*g-1*s4*A2     | None     |
| {#k_e}   | Coulomb const.    | 8.9875517923(14)e9*m3*kg-1*s-4*A-2  | None     |
| {#H_0}   | Hubble const.     | 67.36(54)*3.24076e-20*s-1           | None     |
| {#L_sol} | Solar luminosity  | 3.828e26*m2*kg*s-3                  | None     |
| {#M_sol} | Solar mass        | 1.988475(92)e30*kg                  | None     |
| {#R_sol} | Solar radius      | 6.9566(14)e8*m                      | None     |

## 5 SI: International System of Units

Select this system with the leading `SI_` prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

### 5.1 Derived units and aliases

| Symbol | Meaning           | Definition or reference | Prefixes                     |
|--------|-------------------|-------------------------|------------------------------|
| au     | astr. unit        | 149597.870691*Mm        | None                         |
| AU     | astr. unit        | au                      | None                         |
| ly     | light-year        | {c}*yr_j                | k, M, G                      |
| pc     | parsec            | 3.0857e16*m             | k, M, G, T                   |
| Ao     | Angstrom          | 1e-10*m                 | m, k                         |
| u      | atomic mass unit  | g/{N_0}                 | None                         |
| amu    | atomic mass unit  | u                       | None                         |
| Da     | Dalton            | u                       | None                         |
| t      | tonne             | 1e3*kg                  | k, m, G                      |
| ct     | carat             | 200*mg                  | None                         |
| min    | minute            | 60*s                    | None                         |
| h      | hour              | 60*min                  | None                         |
| day    | day               | 24*h                    | None                         |
| yr_t   | tropical year     | 365.24219*day           | k, m, G                      |
| yr_j   | Julian year       | 365.25*day              | k, m, G                      |
| yr_g   | Gregorian year    | 365.2425*day            | k, m, G                      |
| yr     | year              | yr_j                    | k, m, G                      |
| degR   | degree Rankine    | 5/9*K                   | None                         |
| deg    | angle degree      | 2*{pi}*rad/360          | None                         |
| '      | angle minute      | deg/60                  | None                         |
| ''     | angle second      | '/60                    | None                         |
| ar     | are               | 100*m2                  | c, d, da, h                  |
| b      | barn              | 1e-28*m2                | M, k, m, u, n, p, f, a, z, y |
| l      | liter             | dm3                     | All                          |
| L      | liter             | l                       | All                          |
| J      | Joule             | N*m                     | All                          |
| eV     | electronvolt      | {e}*V                   | All                          |
| erg    | erg               | dyn*cm                  | None                         |
| cal    | calorie           | 4.184*J                 | k, M                         |
| Cal    | Calorie           | kcal                    | None                         |
| Ha     | Hartree           | 4.3597447222071e-18*J   | k, M                         |
| E_h    | Hartree           | Ha                      | k, M                         |
| Pa     | Pascal            | N/m2                    | All                          |
| atm    | atm. pressure     | 101325*Pa               | None                         |
| bar    | bar               | 100*kPa                 | M, k, d, c, m                |
| Ba     | Barye             | 0.1*Pa                  | None                         |
| N      | Newton            | kg*m/s2                 | All                          |
| dyn    | dyne              | g*cm/s2                 | All                          |
| Gy     | Gray              | J/kg                    | None                         |
| Sv     | Sivert            | J/kg                    | None                         |
| T      | Tesla             | Wb/m2                   | All                          |
| G      | Gauss             | 1e-4*T                  | All                          |
| Hz     | Hertz             | s-1                     | All                          |
| Bq     | Becquerel         | s-1                     | None                         |
| W      | Watt              | J/s                     | All                          |
| hp     | horse power       | 745.7*W                 | None                         |
| kph    | kilometer p. hour | km/h                    | None                         |
| kn     | knot              | 0.514444*m/s            | None                         |
| Mx     | Maxwell           | 1e-8*Wb                 | None                         |

| Symbol   | Meaning                     | Definition or reference                                            | Prefixes |
|----------|-----------------------------|--------------------------------------------------------------------|----------|
| Wb       | Weber                       | $V \cdot s$                                                        | None     |
| P        | Poise                       | $g / (cm \cdot s)$                                                 | c        |
| St       | Stokes                      | $cm^2 / s$                                                         | c        |
| Ka       | Kayser                      | $cm^{-1}$                                                          | None     |
| D        | Debye                       | $3.33564e-30 \cdot C \cdot m$                                      | All      |
| Oe       | Oersted                     | $1e3 \cdot A / (4 \cdot \{pi\} \cdot m)$                           | None     |
| Gal      | Gal                         | $cm / s^2$                                                         | None     |
| Bi       | Biot                        | $10 \cdot A$                                                       | None     |
| Rad      | radiation dose              | $0.01 \cdot Gy$                                                    | None     |
| sr       | steradian                   | $rad^2$                                                            | None     |
| lm       | lumen                       | $cd \cdot sr$                                                      | None     |
| C        | coulomb                     | $A \cdot s$                                                        | All      |
| H        | Henry                       | $Wb / A$                                                           | All      |
| Ohm      | Ohm                         | $V / A$                                                            | All      |
| V        | Volt                        | $J / C$                                                            | All      |
| lx       | lux                         | $lm / m^2$                                                         | None     |
| F        | Farad                       | $C / V$                                                            | All      |
| S        | Siemens                     | $Ohm^{-1}$                                                         | All      |
| kat      | katal                       | $mol / s$                                                          | All      |
| dpt      | diopetre                    | $1 / m$                                                            | All      |
| Mo       | seismic moment              | $N \cdot m$                                                        | All      |
| %        | percent                     | $1e-2$                                                             | None     |
| permille | promile                     | $1e-3$                                                             | None     |
| ppth     | promile                     | $1e-3$                                                             | None     |
| ppm      | parts per million           | $1e-6$                                                             | None     |
| ppb      | parts per billion           | $1e-9$                                                             | None     |
| Cel      | degree Celsius              | K (temperature reference;<br>offset conversion)                    | None     |
| degF     | degree Fahrenheit           | K (temperature reference;<br>offset conversion)                    | None     |
| PR       | Power ratio                 | 1                                                                  | None     |
| AR       | Amplitude ratio             | 1                                                                  | None     |
| Np       | Nepers                      | 1 (logarithmic reference;<br>special conversion)                   | c, d     |
| B        | Bel                         | 1 (logarithmic reference;<br>special conversion)                   | d        |
| Bm       | bel-milliwatt               | mW (logarithmic reference;<br>special conversion)                  | d        |
| BmW      | bel-milliwatt               | mW (logarithmic reference;<br>special conversion)                  | d        |
| BW       | bel-watt                    | W (logarithmic reference;<br>special conversion)                   | d        |
| BV       | bel-volt                    | V (logarithmic reference;<br>special conversion)                   | d        |
| BuV      | bel-microvolt               | uV (logarithmic reference;<br>special conversion)                  | d        |
| BA       | bel-amps                    | A (logarithmic reference;<br>special conversion)                   | d        |
| BuA      | bel-microamps               | uA (logarithmic reference;<br>special conversion)                  | d        |
| BOhm     | bel-ohms                    | Ohm (logarithmic<br>reference; special<br>conversion)              | d        |
| BSPL     | bel-SPL (Pa)                | Pa (logarithmic reference;<br>special conversion)                  | d        |
| BSIL     | bel-SIL (W/m <sup>2</sup> ) | W/m <sup>2</sup> (logarithmic<br>reference; special<br>conversion) | d        |

| Symbol | Meaning        | Definition or reference                        | Prefixes |
|--------|----------------|------------------------------------------------|----------|
| BSWL   | bel-SWL (W)    | W (logarithmic reference; special conversion)  | d        |
| Mw     | mom. magnitude | Mo (logarithmic reference; special conversion) | None     |

## 5.2 System-specific constants

| Symbol  | Meaning           | Definition or reference | Prefixes |
|---------|-------------------|-------------------------|----------|
| {alpha} | fine str. const.  | {#alpha}                | None     |
| {euler} | Euler's num.      | {#euler}                | None     |
| {N_0}   | Avogadro's num.   | {#N_0}                  | None     |
| {pi}    | pi num.           | {#pi}                   | None     |
| {a_0}   | Bohr radius       | {#a_0}                  | None     |
| {c}     | speed of light    | {#c}                    | None     |
| {e}     | elem. charge      | {#e}                    | None     |
| {eps_0} | permit. of vac.   | {#eps_0}                | None     |
| {G}     | grav. const.      | {#G}                    | None     |
| {g}     | grav. accel.      | {#g}                    | None     |
| {h}     | Planck const.     | {#h}                    | None     |
| {hbar}  | Reduced Pl. con.  | {#hbar}                 | None     |
| {H_0}   | Hubble const.     | {#H_0}                  | None     |
| {k_B}   | Boltzmann const.  | {#k_B}                  | None     |
| {k_e}   | Coulomb const.    | {#k_e}                  | None     |
| {L_sol} | Solar luminosity  | {#L_sol}                | None     |
| {M_sol} | Solar mass        | {#M_sol}                | None     |
| {mu_0}  | permeab. of vac.  | {#mu_0}                 | None     |
| {mu_B}  | Bohr magneton     | {#mu_B}                 | None     |
| {m_e}   | electron mass     | {#m_e}                  | None     |
| {m_p}   | proton mass       | {#m_p}                  | None     |
| {m_n}   | neutron mass      | {#m_n}                  | None     |
| {R_inf} | Rydberg constant  | {#R_inf}                | None     |
| {R_sol} | Solar radius      | {#R_sol}                | None     |
| {sigma} | Stef.Bolt. const. | {#sigma}                | None     |
| {N_A}   | Avogadro's const. | {#N_A}                  | None     |

## 5.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol | Meaning              | Quantity scale   | Conversion factor |
|--------|----------------------|------------------|-------------------|
| <l>    | length               | m                | —                 |
| <m>    | mass                 | g                | —                 |
| <t>    | time                 | s                | —                 |
| <T>    | temperature          | K                | —                 |
| <I>    | current              | A                | —                 |
| <Iv>   | luminous intensity   | cd               | —                 |
| <n>    | amount of substance  | mol              | —                 |
| <the>  | angular displacement | rad              | —                 |
| <v>    | velocity             | m/s              | —                 |
| <a>    | acceleration         | m/s <sup>2</sup> | —                 |
| <jerk> | jerk                 | m/s <sup>3</sup> | —                 |
| <snap> | snap                 | m/s <sup>4</sup> | —                 |
| <crac> | crackle              | m/s <sup>5</sup> | —                 |

| Symbol    | Meaning                     | Quantity scale                                  | Conversion factor |
|-----------|-----------------------------|-------------------------------------------------|-------------------|
| <pop>     | pop                         | m/s <sup>6</sup>                                | —                 |
| <rho>     | density                     | kg/m <sup>3</sup>                               | —                 |
| <rho_A>   | area density                | kg/m <sup>2</sup>                               | —                 |
| <rho_l>   | linear density              | kg/m <sup>1</sup>                               | —                 |
| <rho_E>   | energy density              | J/m <sup>3</sup>                                | —                 |
| <A>       | area                        | m <sup>2</sup>                                  | —                 |
| <V>       | volume                      | m <sup>3</sup>                                  | —                 |
| <p_mom>   | momentum                    | kg*m/s                                          | —                 |
| <I_mom>   | moment of inertia           | kg*m <sup>2</sup>                               | —                 |
| <ome>     | angular velocity            | rad/s                                           | —                 |
| <a_ome>   | angular acceleration        | rad/s <sup>2</sup>                              | —                 |
| <L_ome>   | angular momentum            | kg*m <sup>2</sup> /s                            | —                 |
| <F>       | force                       | N                                               | —                 |
| <freq>    | frequency                   | Hz                                              | —                 |
| <J_i>     | impulse                     | N*s                                             | —                 |
| <Abs>     | absent                      | m*s                                             | —                 |
| <mu_v>    | dynamic viscosity           | Pa*s                                            | —                 |
| <nu_v>    | kinematic viscosity         | m <sup>2</sup> /s                               | —                 |
| <eps_s>   | strain                      | 1                                               | —                 |
| <sig_s>   | stress                      | Pa                                              | —                 |
| <E_Y>     | Young's modulus             | Pa                                              | —                 |
| <k_s>     | spring constant             | N/m                                             | —                 |
| <gam>     | surface tension             | J/m <sup>2</sup>                                | —                 |
| <tau>     | torque                      | N*m                                             | —                 |
| <E>       | energy                      | J                                               | —                 |
| <S>       | entropy                     | J/K                                             | —                 |
| <p>       | pressure                    | Pa                                              | —                 |
| <mu_c>    | chemical potential          | J/mol                                           | —                 |
| <W>       | work                        | J                                               | —                 |
| <P>       | power                       | W                                               | —                 |
| <E_sp>    | specific energy             | J/kg                                            | —                 |
| <C_sp>    | specific heat capacity      | J/(K*kg)                                        | —                 |
| <V_sp>    | specific volume             | m <sup>3</sup> /kg                              | —                 |
| <Q>       | heat                        | J                                               | —                 |
| <C_p>     | heat capacity               | J/K                                             | —                 |
| <phi_Q>   | heat flux density           | W/m <sup>2</sup>                                | —                 |
| <C_mol>   | molar concentration         | mol/m <sup>3</sup>                              | —                 |
| <E_mol>   | molar energy                | J/mol                                           | —                 |
| <c_mol>   | molar heat capacity         | J/(K*mol)                                       | —                 |
| <S_mol>   | molar entropy               | J/(K*mol)                                       | —                 |
| <rr>      | reaction rate               | mol/(m <sup>3</sup> *s)                         | —                 |
| <kap>     | thermal conductance         | W/K                                             | —                 |
| <lam_t>   | thermal conductivity        | W/(m*K)                                         | —                 |
| <R_t>     | thermal resistance          | K/W                                             | —                 |
| <R_lam>   | thermal resistivity         | K*m/W                                           | —                 |
| <Q_vol>   | volumetric flow rate        | m <sup>3</sup> /s                               | —                 |
| <q>       | electric charge             | C                                               | —                 |
| <rho_q>   | electric charge density     | C/m <sup>3</sup>                                | —                 |
| <p_e>     | electric dipole moment      | C*m                                             | —                 |
| <p_qe>    | electric quadrupole moment  | C*m <sup>2</sup>                                | —                 |
| <D_e>     | electric displacement field | C/m <sup>2</sup>                                | —                 |
| <E_e>     | electric field strength     | V/m                                             | —                 |
| <V_e>     | electric field gradient     | V/m <sup>2</sup>                                | —                 |
| <alp_e>   | electric polarizability     | C <sup>2</sup> *m <sup>2</sup> *J <sup>-1</sup> | —                 |
| <alp_he>  | 1st hyperpolarizability     | C <sup>3</sup> *m <sup>3</sup> *J <sup>-2</sup> | —                 |
| <alp_hhe> | 2nd hyperpolarizability     | C <sup>4</sup> *m <sup>4</sup> *J <sup>-3</sup> | —                 |
| <G_e>     | electrical conductance      | S                                               | —                 |
| <sig_e>   | electrical conductivity     | S/m                                             | —                 |
| <phi_e>   | electric potential          | V                                               | —                 |

| Symbol  | Meaning                 | Quantity scale | Conversion factor |
|---------|-------------------------|----------------|-------------------|
| <Phi_E> | electric flux           | V*m            | —                 |
| <R>     | electrical resistance   | Ohm            | —                 |
| <rho_e> | electrical resistivity  | Ohm*m          | —                 |
| <Mag>   | magnetization           | A/m            | —                 |
| <H>     | magnetic field strenght | A/m            | —                 |
| <Phi_M> | magnetic flux           | Wb             | —                 |
| <B>     | magnetic flux density   | T              | —                 |
| <mm>    | magnetic dipole moment  | N*m/T          | —                 |
| <mu>    | permeability            | H/m            | —                 |
| <eps>   | permittivity            | F/m            | —                 |
| <C>     | capacitance             | F              | —                 |
| <J_c>   | current density         | A/m2           | —                 |
| <Z>     | impedance               | Ohm            | —                 |
| <L>     | inductance              | H              | —                 |
| <Rel>   | reluctance              | 1/H            | —                 |
| <P_o>   | optical power           | dpt            | —                 |
| <lam_w> | wave length             | m              | —                 |
| <k_n>   | wave number             | 1/m            | —                 |
| <k_v>   | wave vector             | 1/m            | —                 |
| <E_il>  | illuminance             | lx             | —                 |
| <E_ir>  | irradiance              | W/m2           | —                 |
| <F_l>   | luinous flux            | lm             | —                 |
| <In>    | intensity               | W/m2           | —                 |
| <L_r>   | radiance                | W/(m2*sr)      | —                 |
| <I_r>   | radiant intensity       | W/sr           | —                 |
| <D_a>   | absorbed dose rate      | Gy/s           | —                 |
| <H_de>  | dose equivalent         | Sv             | —                 |
| <D_r>   | radiation dose          | Gy             | —                 |
| <A_r>   | radioactivity           | Bq             | —                 |
| <S_a>   | action                  | J/Hz           | —                 |
| <spin>  | spin                    | kg*m2*s-1      | —                 |

## 6 ESU: Electrostatic CGS units

Select this system with the leading `ESU_` prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

### 6.1 Derived units and aliases

| Symbol | Meaning             | Definition or reference | Prefixes |
|--------|---------------------|-------------------------|----------|
| Gal    | Gal                 | cm/s2                   | None     |
| dyn    | dyne                | g*cm/s2                 | All      |
| Ba     | Barye               | dyn/cm2                 | None     |
| b      | Barye               | Ba                      | None     |
| erg    | erg                 | dyn*cm                  | None     |
| St     | Stokes              | cm2/s                   | c        |
| P      | Poise               | g/(cm*s)                | c        |
| Ka     | Kayser              | cm-1                    | None     |
| Pma    | permeability        | 1e-4*m2                 | None     |
| Rayl   | Rayl                | g/(cm2*s)               | None     |
| radd   | radiation dose      | 100*erg/g               | None     |
| Fr     | franklin            | dyn1:2*cm               | None     |
| statC  | Statcoulomb         | Fr                      | None     |
| esu    | elstat. u.o. charge | Fr                      | None     |
| statA  | Statampere          | Fr/s                    | None     |
| statV  | Statvolt            | erg/Fr                  | None     |

| Symbol  | Meaning   | Definition or reference               | Prefixes |
|---------|-----------|---------------------------------------|----------|
| statOhm | Statohm   | s/cm                                  | None     |
| statF   | Statfarad | cm                                    | None     |
| statH   | Stathenry | s <sup>2</sup> /cm                    | None     |
| statT   | Stattesla | cm <sup>-3</sup> ·2·g <sup>1</sup> ·2 | None     |
| statWb  | Statweber | cm <sup>2</sup> ·g                    | None     |

## 6.2 System-specific constants

| Symbol  | Meaning           | Definition or reference                       | Prefixes |
|---------|-------------------|-----------------------------------------------|----------|
| {m_u}   | atom. mass const. | 1.660539069e-24·g                             | None     |
| {a_0}   | Bohr radius       | 5.291772105e-9·cm                             | None     |
| {k}     | Boltzmann const.  | 1.380649e-16·erg/K                            | None     |
| {m_e}   | electron mass     | 9.10938371e-28·g                              | None     |
| {alpha} | fine-str. const.  | 0.007297352564                                | None     |
| {G}     | gravit. const.    | 6.6743e-8·dyn·cm <sup>2</sup> /g <sup>2</sup> | None     |
| {h}     | Planck const.     | 6.62607015e-27·erg·s                          | None     |
| {hbar}  | reduced Pl. con.  | 1.054571817e-27·erg·s                         | None     |
| {c}     | speed of light    | 2.99792458e10·cm/s                            | None     |
| {pi}    | pi num.           | 3.1415926                                     | None     |
| {e}     | elementary charge | 4.80320471e-10·Fr                             | None     |
| {mu_B}  | Bohr magneton     | 2.780278273e-10·statA·cm <sup>2</sup>         | None     |

## 6.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol  | Meaning                     | Quantity scale        | Conversion factor                     |
|---------|-----------------------------|-----------------------|---------------------------------------|
| <l>     | length                      | cm                    | —                                     |
| <m>     | mass                        | g                     | —                                     |
| <t>     | time                        | s                     | —                                     |
| <Iv>    | luminous intensity          | cd                    | —                                     |
| <n>     | amount of substance         | mol                   | —                                     |
| <the>   | angular displacement        | rad                   | —                                     |
| <v>     | velocity                    | cm/s                  | —                                     |
| <a>     | acceleration                | Gal                   | —                                     |
| <F>     | force                       | dyn                   | —                                     |
| <E>     | energy                      | erg                   | —                                     |
| <P>     | power                       | erg/s                 | —                                     |
| <p>     | pressure                    | Ba                    | —                                     |
| <mu_v>  | dynamic viscosity           | P                     | —                                     |
| <nu_v>  | kinematic viscosity         | St                    | —                                     |
| <k_n>   | wave number                 | Ka                    | —                                     |
| <q>     | electric charge             | statC                 | 1e1/({c}*100)                         |
| <I>     | current                     | statA                 | 1e1/({c}*100)                         |
| <phi_e> | electric potential          | statV                 | 1e-8*({c}*100)                        |
| <E_e>   | electric field strength     | statV/cm              | 1e-6*({c}*100)                        |
| <D_e>   | electric displacement field | statC/cm <sup>2</sup> | 1e5/(4*{pi}*({c}*100))                |
| <p_e>   | electric dipole moment      | statC·cm              | 1e-1/({c}*100)                        |
| <Phi_E> | electric flux               | statC                 | 1e1/(4*{pi}*({c}*100))                |
| <eps>   | permittivity                | 1                     | 1e11/(4*{pi}*({c}*100) <sup>2</sup> ) |
| <R>     | electrical resistance       | statOhm               | ({c}*100) <sup>2</sup> /1e9           |
| <rho_e> | electrical resistivity      | statOhm·cm            | ({c}*100) <sup>2</sup> /1e11          |
| <C>     | capacitance                 | statF                 | 1e9/({c}*100) <sup>2</sup>            |

| Symbol  | Meaning                 | Quantity scale | Conversion factor            |
|---------|-------------------------|----------------|------------------------------|
| <L>     | inductance              | statH          | $(\{c\}*100)2/1e9$           |
| <B>     | magnetic flux density   | statT          | $1e-4*(\{c\}*100)$           |
| <H>     | magnetic field strenght | statA/cm       | $1e3/(4*\{pi\}*(\{c\}*100))$ |
| <mm>    | magnetic dipole moment  | statA*cm2      | $1e-3/(\{c\}*100)$           |
| <Phi_M> | magnetic flux           | statWb         | $(\{c\}*100)*1e-8$           |
| <mu>    | permeability            | s2/cm2         | $4*\{pi\}*(\{c\}*100)2*1e-7$ |

## 7 EMU: Electromagnetic CGS units

Select this system with the leading `EMU_` prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

### 7.1 Derived units and aliases

| Symbol | Meaning        | Definition or reference | Prefixes |
|--------|----------------|-------------------------|----------|
| Gal    | Gal            | cm/s2                   | None     |
| dyn    | dyne           | g*cm/s2                 | All      |
| Ba     | Barye          | dyn/cm2                 | None     |
| b      | Barye          | Ba                      | None     |
| erg    | erg            | dyn*cm                  | None     |
| St     | Stokes         | cm2/s                   | c        |
| P      | Poise          | g/(cm*s)                | c        |
| Ka     | Kayser         | cm-1                    | None     |
| Pma    | permeability   | 1e-4*m2                 | None     |
| Rayl   | Rayl           | g/(cm2*s)               | None     |
| radd   | radiation dose | 100*erg/g               | None     |
| G      | Gauss          | cm-1:2*g1:2*s-1         | None     |
| Oe     | Oersted        | g1:2/(cm1:2*s)          | None     |
| Gb     | Gilbert        | Oe*cm                   | None     |
| Mx     | Maxwell        | dyn1:2*cm2              | None     |
| Bi     | biot           | dyn1:2                  | None     |
| abA    | Abampere       | Bi                      | None     |
| abC    | Abcoulomb      | Bi*s                    | None     |
| aC     | Abcoulomb      | abC                     | None     |
| abV    | Abvolt         | g1:2*cm3:2*s-2          | None     |
| abOhm  | Abohm          | cm*s                    | None     |
| abF    | Abfarad        | cm-1*s2                 | None     |
| abH    | Abhenry        | cm                      | None     |

### 7.2 System-specific constants

| Symbol  | Meaning           | Definition or reference | Prefixes |
|---------|-------------------|-------------------------|----------|
| {m_u}   | atom. mass const. | 1.660539069e-24*g       | None     |
| {a_0}   | Bohr radius       | 5.291772105e-9*cm       | None     |
| {k}     | Boltzmann const.  | 1.380649e-16*erg/K      | None     |
| {m_e}   | electron mass     | 9.10938371e-28*g        | None     |
| {alpha} | fine-str. const.  | 0.007297352564          | None     |
| {G}     | gravit. const.    | 6.6743e-8*dyn*cm2/g2    | None     |
| {h}     | Planck const.     | 6.62607015e-27*erg*s    | None     |
| {hbar}  | reduced Pl. con.  | 1.054571817e-27*erg*s   | None     |
| {c}     | speed of light    | 2.99792458e10*cm/s      | None     |
| {pi}    | pi num.           | 3.1415926               | None     |
| {mu_B}  | Bohr magneton     | 9.274010066e-21*erg/G   | None     |
| {e}     | elementary charge | 1.602176634e-20*abC     | None     |

### 7.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol  | Meaning                     | Quantity scale | Conversion factor |
|---------|-----------------------------|----------------|-------------------|
| <l>     | length                      | cm             | —                 |
| <m>     | mass                        | g              | —                 |
| <t>     | time                        | s              | —                 |
| <Iv>    | luminous intensity          | cd             | —                 |
| <n>     | amount of substance         | mol            | —                 |
| <the>   | angular displacement        | rad            | —                 |
| <v>     | velocity                    | cm/s           | —                 |
| <a>     | acceleration                | Gal            | —                 |
| <F>     | force                       | dyn            | —                 |
| <E>     | energy                      | erg            | —                 |
| <P>     | power                       | erg/s          | —                 |
| <p>     | pressure                    | Ba             | —                 |
| <mu_v>  | dynamic viscosity           | P              | —                 |
| <nu_v>  | kinematic viscosity         | St             | —                 |
| <k_n>   | wave number                 | Ka             | —                 |
| <B>     | magnetic flux density       | G              | 1e-4              |
| <H>     | magnetic field strenght     | Oe             | 1e3/(4*{pi})      |
| <mm>    | magnetic dipole moment      | erg/G          | 1e-3              |
| <Phi_M> | magnetic flux               | Mx             | 1e-8              |
| <mu>    | permeability                | 1              | 4*{pi}*1e-7       |
| <q>     | electric charge             | abC            | 1e1               |
| <I>     | current                     | abA            | 1e1               |
| <phi_e> | electric potential          | abV            | 1e-8              |
| <E_e>   | electric field strength     | abV/cm         | 1e-6              |
| <D_e>   | electric displacement field | abC/cm2        | 1e5/(4*{pi})      |
| <p_e>   | electric dipole moment      | abC*cm         | 1e-1              |
| <Phi_E> | electric flux               | abC            | 1e1/(4*{pi})      |
| <eps>   | permittivity                | s2/cm2         | 1e11/(4*{pi})     |
| <R>     | electrical resistance       | abOhm          | 1e-9              |
| <rho_e> | electrical resistivity      | abOhm*cm       | 1e-11             |
| <C>     | capacitance                 | abF            | 1e9               |
| <L>     | inductance                  | abH            | 1e-9              |

## 8 GU: Gaussian CGS units

Select this system with the leading GU\_ prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

### 8.1 Derived units and aliases

| Symbol | Meaning | Definition or reference | Prefixes |
|--------|---------|-------------------------|----------|
| Gal    | Gal     | cm/s2                   | None     |
| dyn    | dyne    | g*cm/s2                 | All      |
| Ba     | Barye   | dyn/cm2                 | None     |
| b      | Barye   | Ba                      | None     |
| erg    | erg     | dyn*cm                  | None     |
| St     | Stokes  | cm2/s                   | c        |
| P      | Poise   | g/(cm*s)                | c        |
| Ka     | Kayser  | cm-1                    | None     |

| Symbol  | Meaning             | Definition or reference                         | Prefixes |
|---------|---------------------|-------------------------------------------------|----------|
| Pma     | permeability        | $1\text{e-}4\cdot\text{m}^2$                    | None     |
| Rayl    | Rayl                | $\text{g}/(\text{cm}^2\cdot\text{s})$           | None     |
| radd    | radiation dose      | $100\cdot\text{erg}/\text{g}$                   | None     |
| Fr      | franklin            | $\text{dyn}^1:2\cdot\text{cm}$                  | None     |
| statC   | Statcoulomb         | Fr                                              | None     |
| esu     | elstat. u.o. charge | Fr                                              | None     |
| statA   | Statampere          | $\text{Fr}/\text{s}$                            | None     |
| statV   | Statvolt            | $\text{erg}/\text{Fr}$                          | None     |
| statOhm | Statohm             | $\text{s}/\text{cm}$                            | None     |
| statF   | Statfarad           | cm                                              | None     |
| statH   | Stathenry           | $\text{s}^2/\text{cm}$                          | None     |
| G       | Gauss               | $\text{cm-}1:2\cdot\text{g}^1:2\cdot\text{s-}1$ | None     |
| Oe      | Oersted             | $\text{g}^1:2/(\text{cm}^1:2\cdot\text{s})$     | None     |
| Gb      | Gilbert             | $0\text{e}\cdot\text{cm}$                       | None     |
| Mx      | Maxwell             | $\text{dyn}^1:2\cdot\text{cm}^2$                | None     |

## 8.2 System-specific constants

| Symbol  | Meaning           | Definition or reference                                      | Prefixes |
|---------|-------------------|--------------------------------------------------------------|----------|
| {m_u}   | atom. mass const. | $1.660539069\text{e-}24\cdot\text{g}$                        | None     |
| {a_0}   | Bohr radius       | $5.291772105\text{e-}9\cdot\text{cm}$                        | None     |
| {k}     | Boltzmann const.  | $1.380649\text{e-}16\cdot\text{erg}/\text{K}$                | None     |
| {m_e}   | electron mass     | $9.10938371\text{e-}28\cdot\text{g}$                         | None     |
| {alpha} | fine-str. const.  | $0.007297352564$                                             | None     |
| {G}     | gravit. const.    | $6.6743\text{e-}8\cdot\text{dyn}\cdot\text{cm}^2/\text{g}^2$ | None     |
| {h}     | Planck const.     | $6.62607015\text{e-}27\cdot\text{erg}\cdot\text{s}$          | None     |
| {hbar}  | reduced Pl. con.  | $1.054571817\text{e-}27\cdot\text{erg}\cdot\text{s}$         | None     |
| {c}     | speed of light    | $2.99792458\text{e}10\cdot\text{cm}/\text{s}$                | None     |
| {pi}    | pi num.           | $3.1415926$                                                  | None     |
| {e}     | elementary charge | $4.80320471\text{e-}10\cdot\text{Fr}$                        | None     |
| {mu_B}  | Bohr magneton     | $9.274010066\text{e-}21\cdot\text{erg}/\text{G}$             | None     |

## 8.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol | Meaning              | Quantity scale | Conversion factor             |
|--------|----------------------|----------------|-------------------------------|
| <l>    | length               | cm             | —                             |
| <m>    | mass                 | g              | —                             |
| <t>    | time                 | s              | —                             |
| <Iv>   | luminous intensity   | cd             | —                             |
| <n>    | amount of substance  | mol            | —                             |
| <the>  | angular displacement | rad            | —                             |
| <v>    | velocity             | cm/s           | —                             |
| <a>    | acceleration         | Gal            | —                             |
| <F>    | force                | dyn            | —                             |
| <E>    | energy               | erg            | —                             |
| <P>    | power                | erg/s          | —                             |
| <p>    | pressure             | Ba             | —                             |
| <mu_v> | dynamic viscosity    | P              | —                             |
| <nu_v> | kinematic viscosity  | St             | —                             |
| <k_n>  | wave number          | Ka             | —                             |
| <q>    | electric charge      | statC          | $1\text{e}1/(\{c\}\cdot 100)$ |

| Symbol  | Meaning                     | Quantity scale | Conversion factor                      |
|---------|-----------------------------|----------------|----------------------------------------|
| <I>     | current                     | statA          | $1\text{e}1/(\{c\}*100)$               |
| <phi_e> | electric potential          | statV          | $1\text{e}-8*(\{c\}*100)$              |
| <E_e>   | electric field strength     | statV/cm       | $1\text{e}-6*(\{c\}*100)$              |
| <D_e>   | electric displacement field | statC/cm2      | $1\text{e}5/(4*\{pi\}*(\{c\}*100))$    |
| <p_e>   | electric dipole moment      | statC*cm       | $1\text{e}-1/(\{c\}*100)$              |
| <Phi_E> | electric flux               | statC          | $1\text{e}1/(4*\{pi\}*(\{c\}*100))$    |
| <eps>   | permittivity                | 1              | $1\text{e}11/(4*\{pi\}*(\{c\}*100)^2)$ |
| <R>     | electrical resistance       | statOhm        | $(\{c\}*100)^2/1\text{e}9$             |
| <rho_e> | electrical resistivity      | statOhm*cm     | $(\{c\}*100)^2/1\text{e}11$            |
| <C>     | capacitance                 | statF          | $1\text{e}9/(\{c\}*100)^2$             |
| <L>     | inductance                  | statH          | $(\{c\}*100)^2/1\text{e}9$             |
| <B>     | magnetic flux density       | G              | $1\text{e}-4$                          |
| <H>     | magnetic field strength     | Oe             | $1\text{e}3/(4*\{pi\})$                |
| <mm>    | magnetic dipole moment      | erg/G          | $1\text{e}-3$                          |
| <Phi_M> | magnetic flux               | Mx             | $1\text{e}-8$                          |
| <mu>    | permeability                | 1              | $4*\{pi\}*1\text{e}-7$                 |

## 9 IU: Imperial units

Select this system with the leading IU\_ prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

### 9.1 Derived units and aliases

| Symbol | Meaning            | Definition or reference          | Prefixes |
|--------|--------------------|----------------------------------|----------|
| twip   | twip               | $17.6388888\text{e}-6*\text{m}$  | None     |
| th     | thou               | $25.4\text{e}-6*\text{m}$        | None     |
| mil    | mil                | th                               | None     |
| in     | inch               | $0.0254*\text{m}$                | None     |
| ''     | inch               | in                               | None     |
| li     | link               | $0.201168*\text{m}$              | None     |
| ft     | foot               | $0.3048*\text{m}$                | None     |
| '      | foot               | ft                               | None     |
| yd     | yard               | $0.9144*\text{m}$                | None     |
| rd     | rod/pole/perch     | $5.0292*\text{m}$                | None     |
| ch     | chain              | $20.1168*\text{m}$               | None     |
| fur    | furlong            | $201.168*\text{m}$               | None     |
| mi     | mile               | $1609.344*\text{m}$              | None     |
| lea    | league             | $4828.032*\text{m}$              | None     |
| le     | league             | lea                              | None     |
| ftm    | fathom             | $1.8288*\text{m}$                | None     |
| cb     | cable              | $219.456*\text{m}$               | None     |
| nmi    | nautical mile      | $1852*\text{m}$                  | None     |
| NM     | nautical mile      | nmi                              | None     |
| ac     | acre               | $4046.8564224*\text{m}^2$        | None     |
| gr     | grain              | $64.79891\text{e}-3*\text{g}$    | None     |
| dr     | drachm/dram        | $1.771845195*\text{g}$           | None     |
| oz     | ounce              | $28.349523125*\text{g}$          | None     |
| lb     | pound              | $453.59237*\text{g}$             | None     |
| cwt    | long hundredweight | $50.80234544\text{e}3*\text{g}$  | None     |
| ton    | long ton           | $1.0160469088\text{e}6*\text{g}$ | None     |
| slug   | slug               | $14.59390294\text{e}3*\text{g}$  | None     |
| min    | minute             | $60*\text{s}$                    | None     |
| h      | hour               | $60*\text{min}$                  | None     |
| day    | day                | $24*\text{h}$                    | None     |
| yr_t   | tropical year      | $365.24219*\text{day}$           | k, m, G  |

| Symbol | Meaning           | Definition or reference                         | Prefixes |
|--------|-------------------|-------------------------------------------------|----------|
| yr_j   | Julian year       | 365.25*day                                      | k, m, G  |
| yr_g   | Gregorian year    | 365.2425*day                                    | k, m, G  |
| yr     | year              | yr_j                                            | k, m, G  |
| degR   | degree Rankine    | 5/9*K                                           | None     |
| mph    | miles per hour    | mi/h                                            | None     |
| lbf    | pound-force       | 4.448222e3*g*m*s-2                              | None     |
| Cel    | degree Celsius    | K (temperature reference;<br>offset conversion) | None     |
| degF   | degree Fahrenheit | K (temperature reference;<br>offset conversion) | None     |
| bc     | barleycorn        | 8.4667e-3*m                                     | None     |
| hh     | hand              | 0.1016*m                                        | None     |
| sqrd   | sq. rod/po./per.  | 25.29285264*m2                                  | None     |
| ro     | rood              | 1011.7141056*m2                                 | None     |
| sqmi   | square mile       | 2589988.110336*m2                               | None     |
| minim  | minim             | 59.1938802083e-9*m3                             | None     |
| fls    | fluid scruple     | 1.18387760416e-6*m3                             | None     |
| fldr   | fluid drachm      | 3.5516328125e-6*m3                              | None     |
| floz   | fluid ounce       | 28.4130625e-6*m3                                | None     |
| gi     | gill              | 142.0653125e-6*m3                               | None     |
| pt     | pint              | 568.26125e-6*m3                                 | None     |
| qt     | quart             | 1136.5225e-6*m3                                 | None     |
| gal    | gallon            | 4.54609e-3*m3                                   | None     |
| bblo   | oil barrel        | 0.16365924*m3                                   | None     |
| st     | stone             | 6.35029318e3*g                                  | None     |
| qr     | quarter           | 12.70058636e3*g                                 | None     |

## 9.2 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol | Meaning              | Quantity scale | Conversion factor |
|--------|----------------------|----------------|-------------------|
| <l>    | length               | ft             | —                 |
| <m>    | mass                 | lb             | —                 |
| <t>    | time                 | s              | —                 |
| <Iv>   | luminous intensity   | cd             | —                 |
| <n>    | amount of substance  | mol            | —                 |
| <the>  | angular displacement | rad            | —                 |
| <V>    | volume               | gal            | —                 |
| <A>    | area                 | ac             | —                 |
| <v>    | velocity             | mph            | —                 |

## 10 US: United States customary units

Select this system with the leading **US\_** prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

### 10.1 Derived units and aliases

| Symbol | Meaning | Definition or reference | Prefixes |
|--------|---------|-------------------------|----------|
| twip   | twip    | 17.6388888e-6*m         | None     |
| th     | thou    | 25.4e-6*m               | None     |

| Symbol  | Meaning            | Definition or reference                         | Prefixes |
|---------|--------------------|-------------------------------------------------|----------|
| mil     | mil                | th                                              | None     |
| in      | inch               | 0.0254*m                                        | None     |
| ''      | inch               | in                                              | None     |
| li      | link               | 0.201168*m                                      | None     |
| ft      | foot               | 0.3048*m                                        | None     |
| '       | foot               | ft                                              | None     |
| yd      | yard               | 0.9144*m                                        | None     |
| rd      | rod/pole/perch     | 5.0292*m                                        | None     |
| ch      | chain              | 20.1168*m                                       | None     |
| fur     | furlong            | 201.168*m                                       | None     |
| mi      | mile               | 1609.344*m                                      | None     |
| lea     | league             | 4828.032*m                                      | None     |
| le      | league             | lea                                             | None     |
| ftm     | fathom             | 1.8288*m                                        | None     |
| cb      | cable              | 219.456*m                                       | None     |
| nmi     | nautical mile      | 1852*m                                          | None     |
| NM      | nautical mile      | nmi                                             | None     |
| ac      | acre               | 4046.8564224*m2                                 | None     |
| gr      | grain              | 64.79891e-3*g                                   | None     |
| dr      | drachm/dram        | 1.771845195*g                                   | None     |
| oz      | ounce              | 28.349523125*g                                  | None     |
| lb      | pound              | 453.59237*g                                     | None     |
| cwt     | long hundredweight | 50.80234544e3*g                                 | None     |
| ton     | long ton           | 1.0160469088e6*g                                | None     |
| slug    | slug               | 14.59390294e3*g                                 | None     |
| min     | minute             | 60*s                                            | None     |
| h       | hour               | 60*min                                          | None     |
| day     | day                | 24*h                                            | None     |
| yr_t    | tropical year      | 365.24219*day                                   | k, m, G  |
| yr_j    | Julian year        | 365.25*day                                      | k, m, G  |
| yr_g    | Gregorian year     | 365.2425*day                                    | k, m, G  |
| yr      | year               | yr_j                                            | k, m, G  |
| degR    | degree Rankine     | 5/9*K                                           | None     |
| mph     | miles per hour     | mi/h                                            | None     |
| lbf     | pound-force        | 4.448222e3*g*m*s-2                              | None     |
| Cel     | degree Celsius     | K (temperature reference;<br>offset conversion) | None     |
| degF    | degree Fahrenheit  | K (temperature reference;<br>offset conversion) | None     |
| p       | point              | 352.778e-6*m                                    | None     |
| P       | pica               | 4.233e-3*m                                      | None     |
| sqft    | sq. survey foot    | 0.09290341*m2                                   | None     |
| sqch    | square chain       | 404.68564224*m2                                 | None     |
| section | section            | 2.589998e6*m2                                   | None     |
| twp     | survey township    | 93.23993e6*m2                                   | None     |
| cuin    | cubic inch         | 16.387064e-6*m3                                 | None     |
| cuft    | cubic foot         | 28.316846592e-3*m3                              | None     |
| cuyd    | cubic yard         | 0.764554857984*m3                               | None     |
| acft    | acre-foot          | 1233.482*m3                                     | None     |
| minim   | minim              | 61.611519921875e-9*m3                           | None     |
| fldr    | fluid drachm       | 3.6966911953125e-6*m3                           | None     |
| tsp     | teaspoon           | 4.92892159375e-6*m3                             | None     |
| tbsp    | tablespoon         | 14.78676478125e-6*m3                            | None     |
| floz    | fluid ounce        | 29.5735295625e-6*m3                             | None     |
| jig     | jig                | 44.36029434375e-6*m3                            | None     |
| gi      | gill               | 118.29411825e-6*m3                              | None     |
| c       | cup                | 236.5882365e-6*m3                               | None     |
| pt      | pint               | 0.473176473e-3*m3                               | None     |
| qt      | quart              | 0.946352946e-3*m3                               | None     |

| Symbol | Meaning          | Definition or reference | Prefixes |
|--------|------------------|-------------------------|----------|
| pot    | pottle           | 1.892705892e-3*m3       | None     |
| gal    | gallon           | 3.785411784e-3*m3       | None     |
| bbl    | liquid barrel    | 119.240471196e-3*m3     | None     |
| bblo   | oil barrel       | 158.987294928e-3*m3     | None     |
| hhd    | hogshead         | 238.480942392e-3*m3     | None     |
| ptd    | dry pint         | 0.55061047135749e-3*m3  | None     |
| qtd    | dry quart        | 1.1012209427149e-3*m3   | None     |
| gald   | dry gallon       | 4.4048837708599e-3*m3   | None     |
| pk     | peck             | 8.8097675417199e-3*m3   | None     |
| bu     | bushel           | 35.239070166879e-3*m3   | None     |
| bbll   | liquid barrel    | 115.62712358400e-3*m3   | None     |
| cwts   | short hundredwe. | 45.359237e3*g           | None     |
| tons   | short ton        | 907.18474e3*g           | None     |
| dwt    | pennyweight      | 1.55517384*g            | None     |
| ozt    | troy ounce       | 31.1034768*g            | None     |
| lbt    | troy pound       | 373.241721*g            | None     |

## 10.2 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol | Meaning              | Quantity scale | Conversion factor |
|--------|----------------------|----------------|-------------------|
| <l>    | length               | ft             | —                 |
| <m>    | mass                 | lb             | —                 |
| <t>    | time                 | s              | —                 |
| <Iv>   | luminous intensity   | cd             | —                 |
| <n>    | amount of substance  | mol            | —                 |
| <the>  | angular displacement | rad            | —                 |
| <V>    | volume               | gal            | —                 |
| <A>    | area                 | ac             | —                 |
| <v>    | velocity             | mph            | —                 |

## 11 AU: Atomic units

Select this system with the leading AU\_ prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

### 11.1 Derived units and aliases

| Symbol | Meaning | Definition or reference  | Prefixes |
|--------|---------|--------------------------|----------|
| E_h    | hartree | {#hbar}2/({#m_e}*{a_0}2) | None     |

### 11.2 System-specific constants

| Symbol | Meaning           | Definition or reference | Prefixes |
|--------|-------------------|-------------------------|----------|
| {hbar} | reduced Pl. con.  | 1                       | None     |
| {e}    | elementary charge | 1                       | None     |
| {m_e}  | electr. rest mass | 1                       | None     |

| Symbol  | Meaning           | Definition or reference                            | Prefixes |
|---------|-------------------|----------------------------------------------------|----------|
| {kap_0} | permittivity      | 1                                                  | None     |
| {a_0}   | Bohr radius       | $\{\#kap\_0\}*\{\#hbar\}^2/(\{\#m\_e\}*\{\#e\}^2)$ | None     |
| {alpha} | fine-str. const.  | 1/137                                              | None     |
| {c}     | speed of light    | $\{\alpha\}-1*\{a\_0\}*E\_h/\{\#hbar\}$            | None     |
| {r_e}   | class. e. radius  | $\{\alpha\}^2*\{a\_0\}$                            | None     |
| {lbar}  | red. Com. wavele. | $\{\alpha\}*\{a\_0\}$                              | None     |
| {m_p}   | proton mass       | $1836*\{\#m\_e\}$                                  | None     |

### 11.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol    | Meaning                    | Quantity scale                    | Conversion factor |
|-----------|----------------------------|-----------------------------------|-------------------|
| <l>       | length                     | {a_0}                             | —                 |
| <m>       | mass                       | {#m_e}                            | —                 |
| <t>       | time                       | {#hbar}/E_h                       | —                 |
| <I>       | current                    | {#e}*E_h/{#hbar}                  | —                 |
| <Iv>      | luminous intensity         | cd                                | —                 |
| <n>       | amount of substance        | mol                               | —                 |
| <the>     | angular displacement       | rad                               | —                 |
| <v>       | velocity                   | $\{a\_0\}*E\_h/\{\#hbar\}$        | —                 |
| <p_mom>   | momentum                   | $\{\#hbar\}/\{a\_0\}$             | —                 |
| <rho_q>   | electric charge density    | $\{\#e\}/\{a\_0\}^3$              | —                 |
| <q>       | electric charge            | {#e}                              | —                 |
| <p_e>     | electric dipole moment     | $\{\#e\}*\{a\_0\}$                | —                 |
| <p_qe>    | electric quadrupole moment | $\{\#e\}*\{a\_0\}^2$              | —                 |
| <phi_e>   | electric potential         | $E\_h/\{\#e\}$                    | —                 |
| <E_e>     | electric field strength    | $E\_h/\{\#e\}*\{a\_0\}$           | —                 |
| <V_e>     | electric field gradient    | $E\_h/\{\#e\}*\{a\_0\}^2$         | —                 |
| <eps>     | permittivity               | {#kap_0}                          | —                 |
| <alp_e>   | electric polarizability    | $\{\#e\}^2*\{a\_0\}^2/E\_h$       | —                 |
| <alp_he>  | 1st hyperpolarizability    | $\{\#e\}^3*\{a\_0\}^3/E\_h^2$     | —                 |
| <alp_hhe> | 2nd hyperpolarizability    | $\{\#e\}^4*\{a\_0\}^4/E\_h^3$     | —                 |
| <B>       | magnetic flux density      | $\{\#hbar\}/(\{\#e\}*\{a\_0\}^2)$ | —                 |
| <mm>      | magnetic dipole moment     | $\{\#hbar\}*\{\#e\}/\{\#m\_e\}$   | —                 |
| <E>       | energy                     | E_h                               | —                 |
| <F>       | force                      | $E\_h/\{a\_0\}$                   | —                 |
| <S_a>     | action                     | {#hbar}                           | —                 |

## 12 PU: Planck units

Select this system with the leading PU\_ prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

### 12.1 Derived units and aliases

This system defines its characteristic scales through the quantity symbols below rather than additional named unit symbols.

### 12.2 System-specific constants

| Symbol | Meaning          | Definition or reference | Prefixes |
|--------|------------------|-------------------------|----------|
| {c}    | speed of light   | 1                       | None     |
| {G}    | grav. const.     | 1                       | None     |
| {hbar} | reduced Pl. con. | 1                       | None     |
| {k_B}  | Boltzmann const. | 1                       | None     |

### 12.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol  | Meaning      | Quantity scale                                   | Conversion factor |
|---------|--------------|--------------------------------------------------|-------------------|
| <l>     | length       | $(\{hbar\} * \{G\} / \{c\}^3) 1:2$               | —                 |
| <m>     | mass         | $(\{hbar\} * \{c\} / \{G\}) 1:2$                 | —                 |
| <t>     | time         | $(\{hbar\} * \{G\} / \{c\}^5) 1:2$               | —                 |
| <T>     | temperature  | $(\{hbar\} * \{c\}^5 * \{k_B\} - 2 / \{G\}) 1:2$ | —                 |
| <A>     | area         | $\{hbar\} * \{G\} / \{c\}^3$                     | —                 |
| <V>     | volume       | $(\{hbar\} * \{G\} * \{c\} - 3) 3:2$             | —                 |
| <p_mom> | momentum     | $(\{hbar\} * \{c\}^3 / \{G\}) 1:2$               | —                 |
| <E>     | energy       | $(\{hbar\} * \{c\}^5 / \{G\}) 1:2$               | —                 |
| <F>     | force        | $\{c\}^4 / \{G\}$                                | —                 |
| <rho>   | density      | $\{c\}^5 / \{hbar\} * \{G\} - 2$                 | —                 |
| <a>     | acceleration | $(\{c\}^7 / \{hbar\} / \{G\}) 1:2$               | —                 |

## 13 SRU: Units for special relativity

Select this system with the leading SRU\_ prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

### 13.1 Derived units and aliases

This system defines its characteristic scales through the quantity symbols below rather than additional named unit symbols.

### 13.2 System-specific constants

| Symbol | Meaning        | Definition or reference | Prefixes |
|--------|----------------|-------------------------|----------|
| {c}    | speed of light | 1                       | None     |

### 13.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol | Meaning | Quantity scale | Conversion factor |
|--------|---------|----------------|-------------------|
| <l>    | length  | m              | —                 |
| <m>    | mass    | kg             | —                 |
| <t>    | time    | m              | {c}-1             |
| <E>    | energy  | kg             | {c}2              |

| Symbol  | Meaning          | Quantity scale | Conversion factor |
|---------|------------------|----------------|-------------------|
| <p_mom> | momentum         | kg             | {#c}              |
| <v>     | velocity         | 1              | {#c}              |
| <L_ome> | angular momentum | kg*m           | {#c}              |
| <F>     | force            | kg/m           | {#c}2             |
| <a>     | acceleration     | m-1            | {#c}2             |
| <rho_E> | energy density   | kg/m3          | {#c}2             |

## 14 GRU: Units for general relativity

Select this system with the leading `GRU_` prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

### 14.1 Derived units and aliases

| Symbol | Meaning      | Definition or reference                                 | Prefixes |
|--------|--------------|---------------------------------------------------------|----------|
| eV     | electronvolt | $1.602176634e-19 \text{ m}^2 \text{ kg} \text{ s}^{-2}$ | M, G, T  |

### 14.2 System-specific constants

| Symbol  | Meaning          | Definition or reference             | Prefixes      |
|---------|------------------|-------------------------------------|---------------|
| {c}     | speed of light   | 1                                   | None          |
| {hbar}  | reduced Pl. con. | 1                                   | None          |
| {eps_0} | permit. of vac.  | 1                                   | None          |
| {k_B}   | Boltzmann const. | 1                                   | None          |
| {pi}    | pi num.          | {#pi}                               | None          |
| {alpha} | fine-str. const. | $\frac{e^2}{4\pi\epsilon_0\hbar c}$ | {#hbar}*{#c}) |
| {e}     | elem. charge     | $(4\pi\epsilon_0\hbar c)\alpha$     | 1:2 None      |

### 14.3 Quantity symbols and conversion factors

<name> denotes the quantity scale in the third column. Where provided, the last column defines |name|, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol  | Meaning          | Quantity scale | Conversion factor |
|---------|------------------|----------------|-------------------|
| <l>     | length           | GeV-1          | {#hbar}*{#c}      |
| <m>     | mass             | GeV            | {#c}-2            |
| <t>     | time             | GeV-1          | {#hbar}           |
| <E>     | energy           | GeV            | —                 |
| <p_mom> | momentum         | GeV            | {#c}-1            |
| <v>     | velocity         | 1              | {#c}              |
| <L_ome> | angular momentum | 1              | {#hbar}           |
| <A>     | area             | GeV-2          | ({#hbar}*{#c})2   |
| <F>     | force            | GeV2           | ({#hbar}*{#c})-1  |
| <rho_E> | energy density   | GeV4           | ({#hbar}*{#c})-3  |
| <q>     | electric charge  | {#e}/{#e}      | —                 |

## 15 GEO: Geometrized units

Select this system with the leading `GEO_` prefix. The shared base-unit symbols are also available. Definitions below are interpreted within this system; aliases and capitalization are significant. Prefix rules follow the shared table legend.

The fixed reference constants are also available.

### 15.1 Derived units and aliases

This system defines its characteristic scales through the quantity symbols below rather than additional named unit symbols.

### 15.2 System-specific constants

| Symbol | Meaning        | Definition or reference | Prefixes |
|--------|----------------|-------------------------|----------|
| {c}    | speed of light | 1                       | None     |
| {G}    | grav. const.   | 1                       | None     |

### 15.3 Quantity symbols and conversion factors

`<name>` denotes the quantity scale in the third column. Where provided, the last column defines `|name|`, the factor used by contextual conversion to relate that scale to SI. A dash means no explicit factor is defined. The factor expressions are evaluated using SI metre-kilogram-second scaling, and their numerical results are stored as dimensionless conversion coefficients.

| Symbol  | Meaning          | Quantity scale | Conversion factor |
|---------|------------------|----------------|-------------------|
| <l>     | length           | m              | —                 |
| <m>     | mass             | m              | {#c}2/{#G}        |
| <t>     | time             | m              | {#c}-1            |
| <E>     | energy           | m              | {#c}4/{#G}        |
| <p_mom> | momentum         | m              | {#c}3/{#G}        |
| <v>     | velocity         | 1              | {#c}              |
| <L_ome> | angular momentum | m2             | {#c}3/{#G}        |
| <F>     | force            | 1              | {#c}4/{#G}        |
| <a>     | acceleration     | m-1            | {#c}2             |
| <rho_E> | energy density   | m-2            | {#c}4/{#G}        |